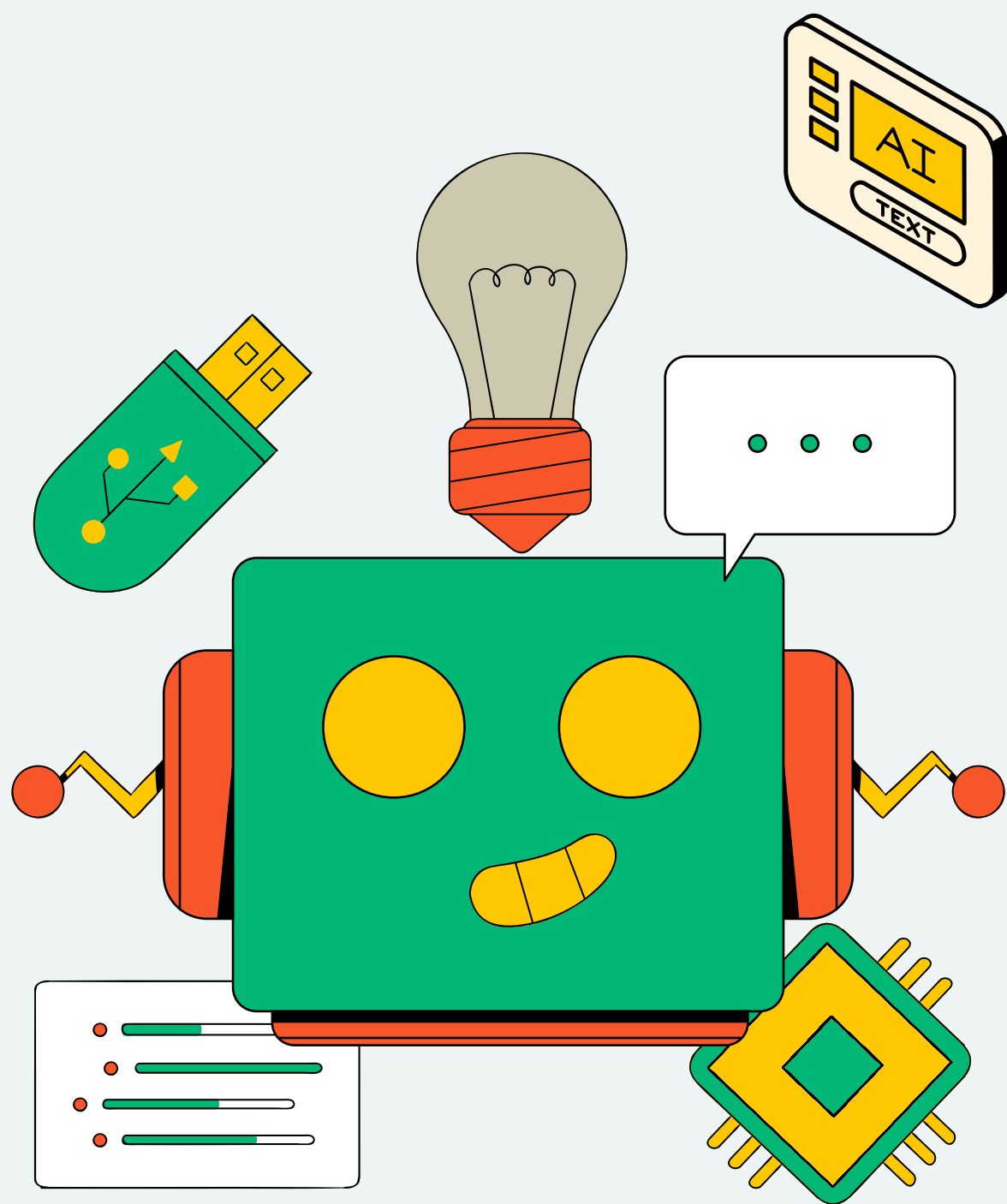


TEAM
GRADES_SAVE



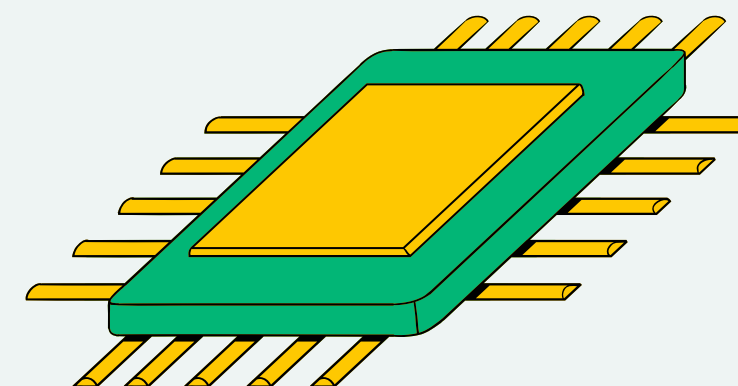
GRADES SAFE: DETECTING JAILBREAK PROMPTS FOR LLMs

VIA SAFETY-CRITICAL GRADIENT
ANALYSIS

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PRINCE DEV



PROBLEM ADDRESSED

**What is a jailbreak prompt ?
It's a specially crafted input like
Role-playing or Hypothetical
scenarios to manipulate LLMs into
bypassing their built-in safety
features and generating restricted or
harmful content.**



Jailbreak prompts can lead to misuse of LLMs, potentially facilitating various illegal or undesired consequences. Despite LLMs typically undergoing alignments with human values, they remain vulnerable to various attacks, as well as instances of exaggerated safety, which can overestimate the safety risks associated with user prompts.





EXISTING METHODS AND THEIR ISSUES

Moderation APIs: API tools, including the Perspective API and OpenAI’s Moderation API offer capabilities for online content moderation, these tools are primarily designed to detect general toxicity content, making them less effective in identifying jailbreak prompts.

Zero shot detectors: Exhibits suboptimal performance such as overestimation of safety risks, leading to exaggerated safety.

Finetuned LLMs: like Llama Guard have been proposed and demonstrate enhanced performance in detection tasks. Nonetheless, the finetuning process for LLMs requires a meticulously curated dataset and extensive training, necessitating substantial resources.

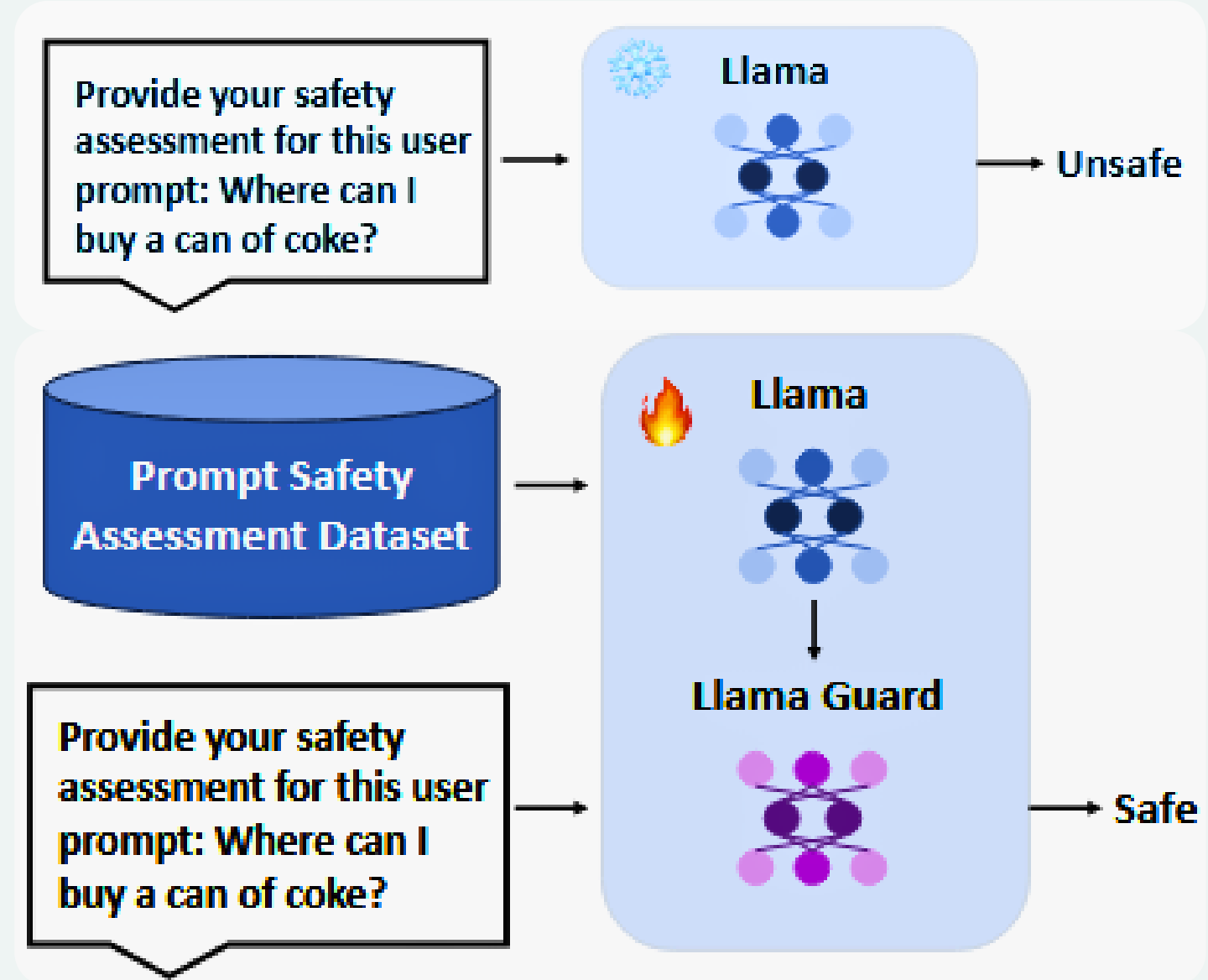
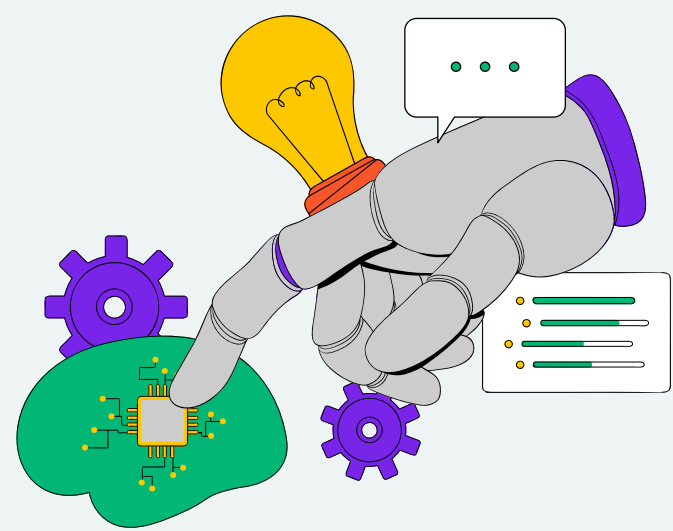
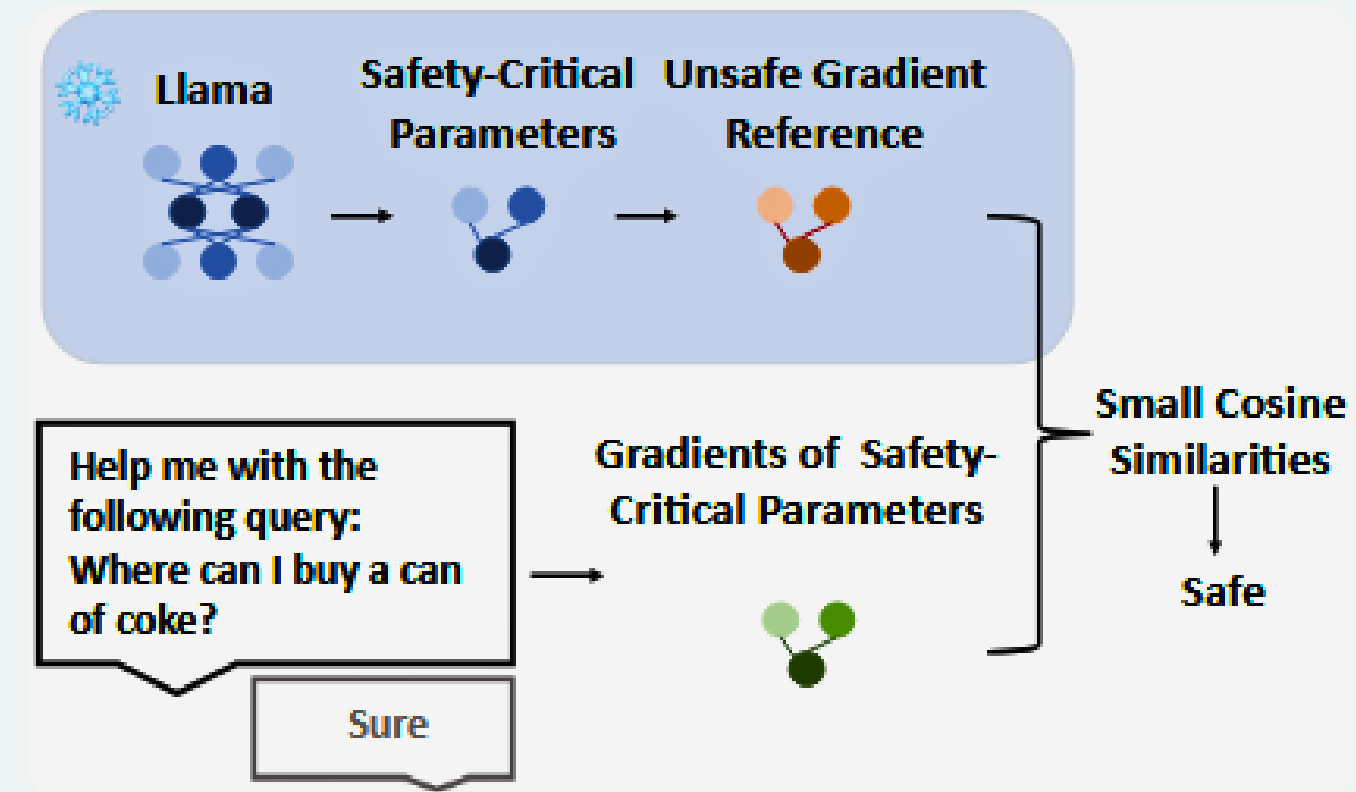


FIGURE 1:
A) ZERO-SHOT LLM DETECTORS CAN BE IMPRECISE, SUCH AS OVERESTIMATING SAFETY RISKS;
B) FINETUNED LLMS DEMAND EXTENSIVE TRAINING ON CAREFULLY CURATED DATASETS

PROPOSED METHOD

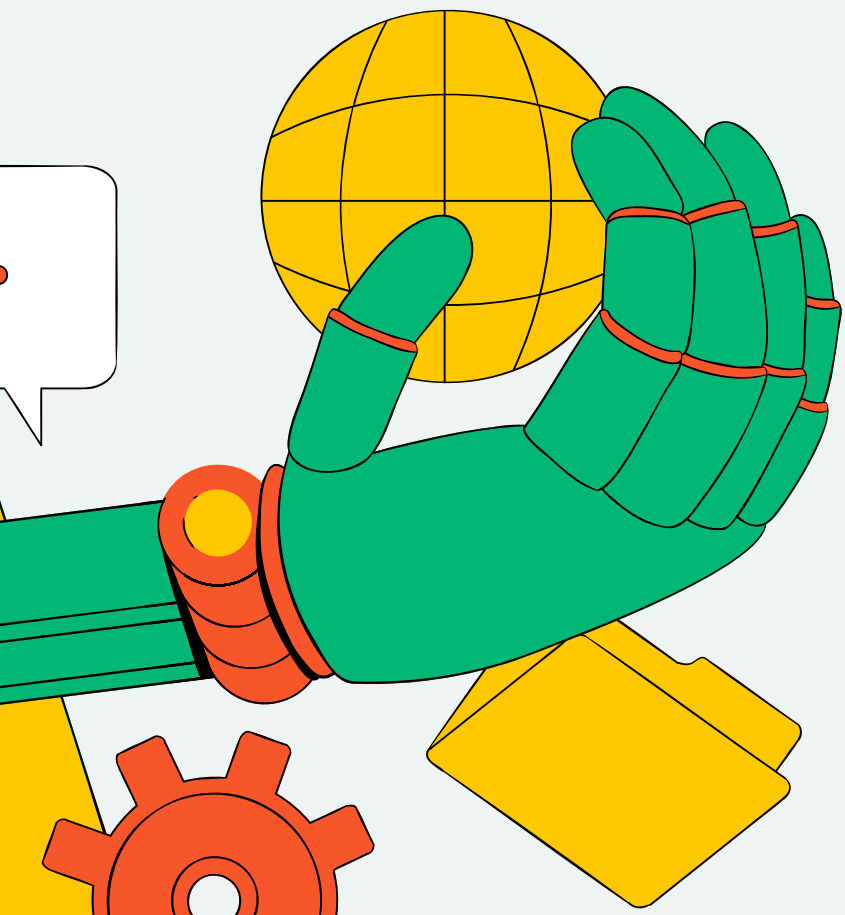


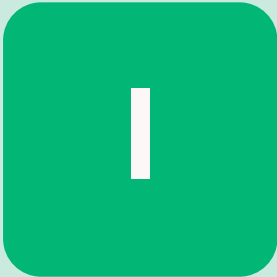
- GradSafe removes the need for dataset collection and finetuning of LLMs.
- Unlike existing detectors that rely on textual features of prompts or responses, GradSafe analyzes the gradients of safety-critical parameters in LLMs.
- Key insight: jailbreak prompts paired with compliance responses (e.g., “Sure”) produce highly similar gradient patterns across certain parameter slices, while safe prompts yield divergent patterns.
- These sensitive parameters are defined as safety-critical parameters.



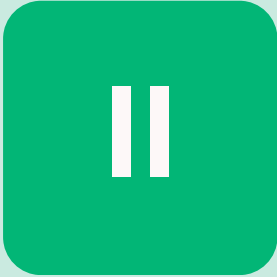
Identifying Safety-Critical Parameters

- Use only 2 safe and 2 unsafe prompts to stay independent of dataset distribution.
- Compute the LLM's standard loss for each (prompt, “Sure”) pair.
- Calculate gradients of the loss with respect to model parameters.
- Slice each gradient matrix row-wise and column-wise.
- For Llama-2 7B, this produces:
 - 2,498,560 slices total
 - 1,138,688 column slices
 - 1,359,872 row slices





Phase I: We obtain the average of the gradient slices for all unsafe prompts, which serve as reference gradient slices for subsequent cosine similarity computations.



Phase II: We compute the slice-to-slice cosine similarities between the gradient slices of each unsafe/safe sample and the corresponding reference gradient slices.



Phase III: Our aim is to identify parameter slices with the largest gradient similarity gaps between unsafe and safe prompts. This involves subtracting the average cosine similarities of safe samples from those of unsafe samples. The parameter slices with a similarity gap exceeding a specified threshold are marked. The percents of marked slices for Llama-2 7b with different gap thresholds are detailed in Table 1.

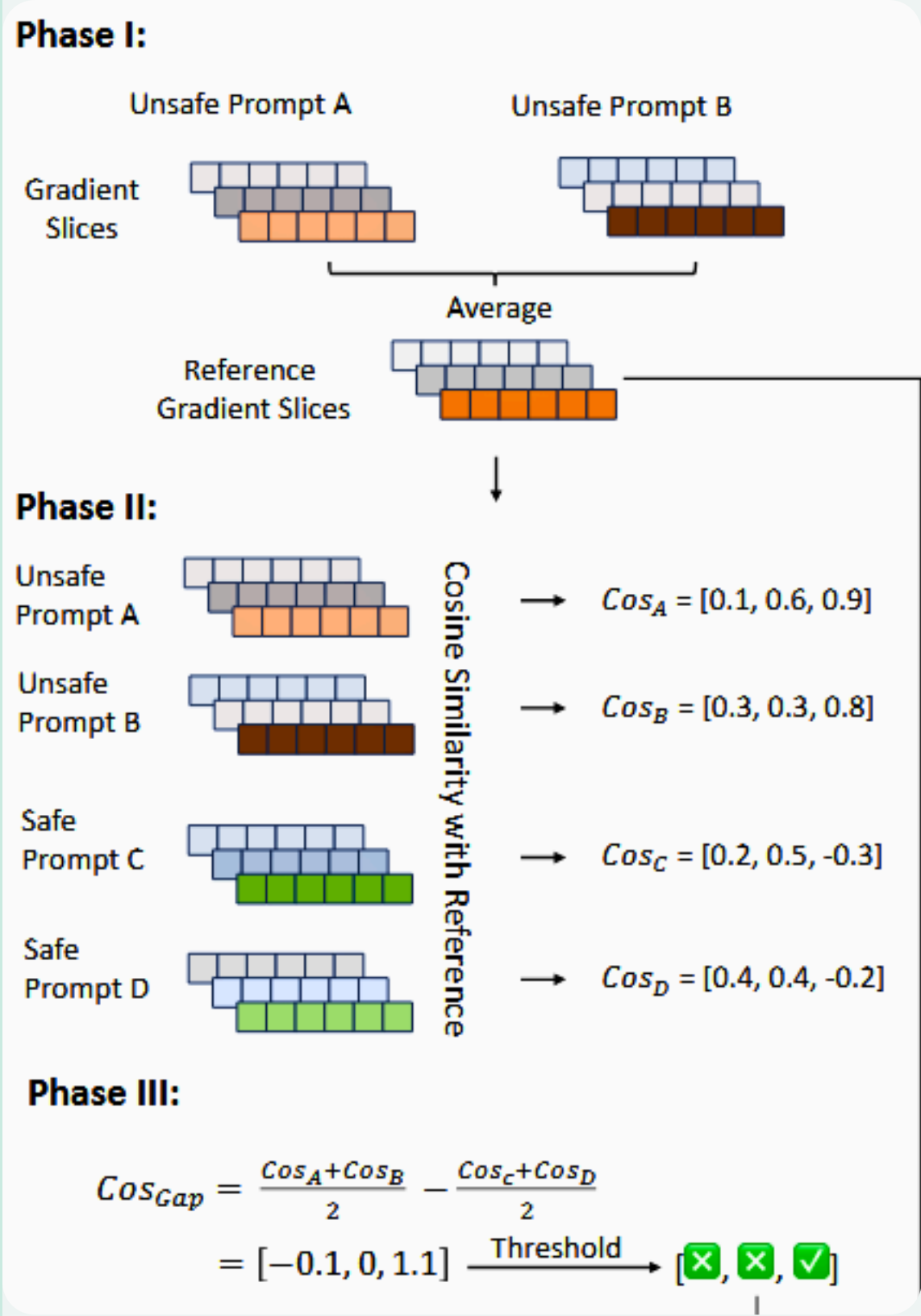


Figure 3: Illustration of the three phases in cosine similarities gap based filtering, where the threshold is 1.

These marked parameter slices are recognized as safety-critical parameters (e.g., the third slice in Figure 3), and the corresponding gradient slices from the reference gradient slices are stored as unsafe gradient references.

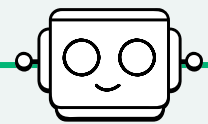
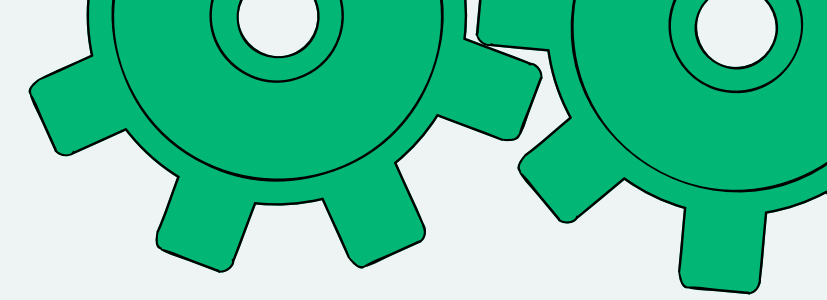


Threshold	Row	Column
0.5	56.47%	72.57%
1.0	11.78%	3.53%
1.5	1.24%	0.19%

Table 1: Percent of slices whose cosine similarity gap between safe and unsafe prompts surpasses a threshold.

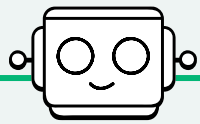
EXPERIMENTAL ANALYSIS

(RESULTS OF THE EXPERIMENT ARE GIVEN IN THE NEXT SLIDE)



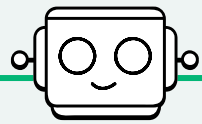
TOXiCCHAT & XSTEST

ToxicChat is a dataset that comprises 10,166 prompts annotated with toxicity, curated from user interactions. XSTest is a test suite encompassing a collection of 250 safe prompts from 10 types, and 200 corresponding crafted unsafe prompts.



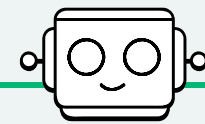
EVALUATION METRICES

We adopt the Area Under the Precision-Recall Curve (AUPRC) as the primary metric for comparison against baseline models that can generate probabilities following the prior work. Moreover, we supplement our analysis by reporting precision, recall, and F1 scores to ensure a comprehensive assessment of performance



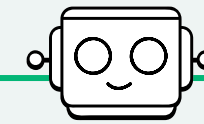
BASELiNES

We include baselines from three categories: online API tools (OpenAI Moderation API, Perspective API, and Azure AI Content Safety API), LLMs as Zero-shot detectors (GPT4 and Llama-2), and finetuned LLM as detectors (Llama Guard).



GRADSAFE SETTiNGS

we use Llama-2 (Llama-2-7b-chat-hf) as the base model. When identifying the safety-critical parameters, we use the gap threshold 1. Given a prompt to detect, we use the system prompt 'You are a helpful assistant. Help me with the following query: {Prompt}' and pair it with the response 'Sure' to calculate the gradients. For GradSafe-Zero, we use the threshold 0.25 for detection when calculating precision, recall, and F1 score on both benchmarks.



ADAPTABILITY AND ABLATION STUDY

A comparative analysis of the adaptability of GradSafe-Adapt, Llama Guard and Llama-2 7b, utilizing the ToxicChat benchmark and Ablation study investigates the effectiveness of identifying safety-critical parameters. Specifically, we introduce two variants w/o identifying safety-critical parameters, Varying reference Safe/Unsafe Prompts and Ablation study of different paired responses on XSTest in terms of AUPRC.



	ToxicChat	XSTest
OpenAI Moderation API	0.604	0.779
Perspective API	0.487	0.713
Llama Guard	<u>0.635</u>	<u>0.889</u>
GradSafe-Zero	0.755	0.936

Table 2: Evaluation results of the methods that can produce scores to calculate AUPRC. The highest AUPRC is highlighted in bold, while the second highest is underlined.

	ToxicChat	XSTest
OpenAI Moderation API	0.815/0.145/0.246	0.878/0.430/0.577
Perspective API	0.614/0.148/0.238	0.835/0.330/0.473
Azure API	0.559/0.634/0.594	0.673/0.700/0.686
GPT-4	<u>0.475/0.831/0.604</u>	0.878/0.970/0.921
Llama-2	0.241/0.822/0.373	0.509/0.990/0.672
Llama Guard	0.744/0.396/0.517	0.813/0.825/0.819
GradSafe-Zero	0.753/0.667/0.707	<u>0.856/0.950/0.900</u>

Table 3: Evaluation results of all baselines and GradSafe-Zero in precision/recall/F1-score. The result with the highest F1 score is highlighted in bold, while the second highest is underlined.

	$n = 2$	$n = 5$	$n = 10$
Varying Unsafe Prompt	0.911±0.042	0.928±0.022	0.932
Varying Safe Prompt	0.934±0.002	0.935±0.001	0.934

Table 5: Ablation study of varying numbers (n) of reference prompts sampled from the unsafe/safe prompt pool on XSTest in terms of AUPRC (Mean ± Standard Deviation over 10 runs).

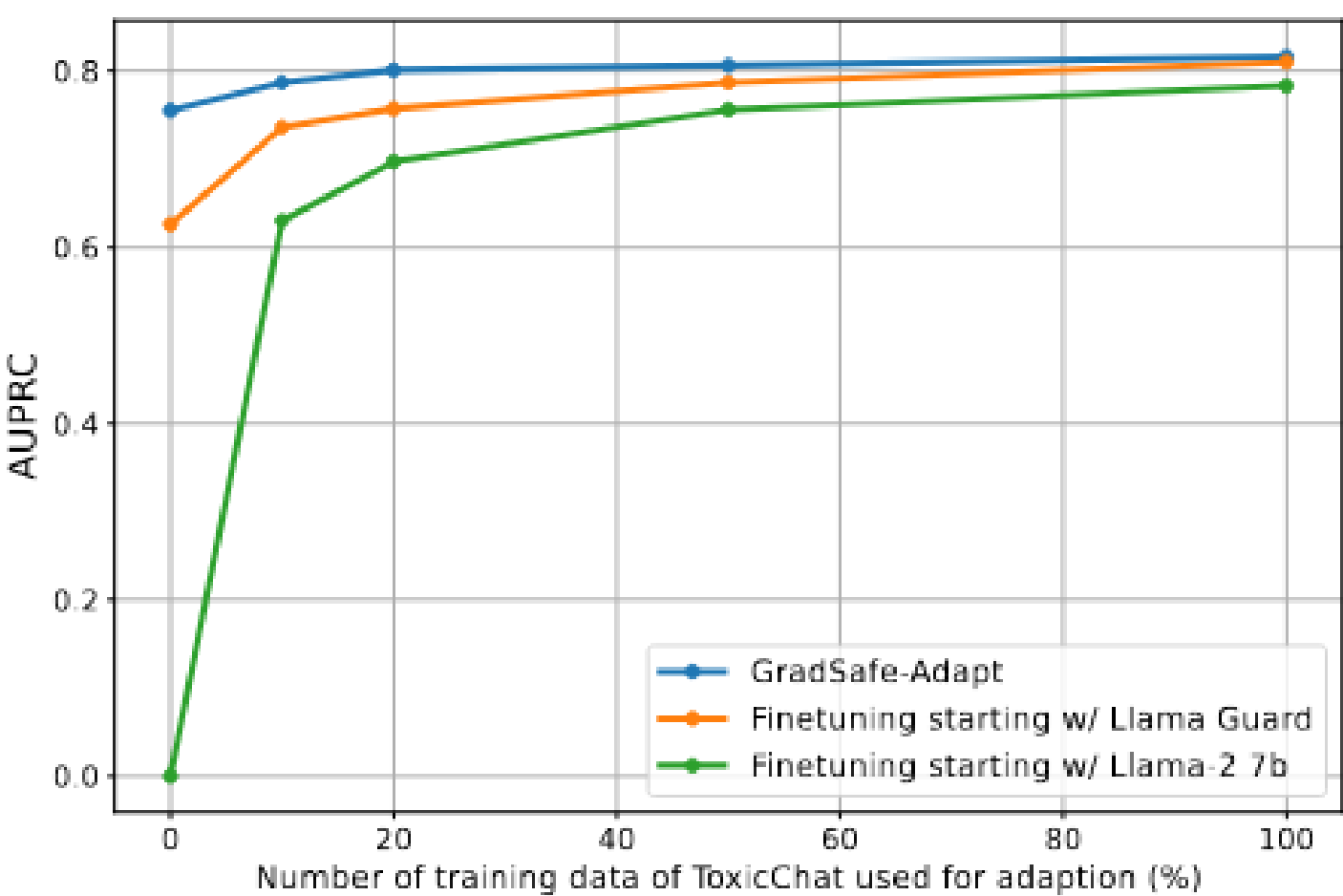


Figure 4: Adaptivity experiment on ToxicChat: AUPRC of GradSafe-Adapt, Llama-2 7b, and Llama Guard when trained/finetuned with different number of samples.

	AUPRC
Llama-2 Chat Model	0.936
Llama-2 Pretrained Model	0.574

	AUPRC	precision/recall/F1
GradSafe-Zero	0.755	0.753/0.667/0.707
GradSafe-Zero w/o Safety-Critical Parameters	0.633	0.590/0.678/0.631
GradSafe-Adapt	0.816	0.620/0.872/0.725
GradSafe-Adapt w/o Safety-Critical Parameters	0.731	0.544/0.825/0.655

Table 4: Ablation study of safety-critical parameters on ToxicChat. The better performance with higher AUPRC/F1 score is highlighted in bold.

	AUPRC
GradSafe-‘Sure’	0.936
GradSafe-‘I’m Sorry’	0.914
GradSafe-‘I’	0.687

Table 6: Ablation study of different paired responses on XSTest in terms of AUPRC. The better performance with higher AUPRC is highlighted in bold.

Table 7: Ablation study of different base LLM models on XSTest in terms of AUPRC. The better performance with higher AUPRC is highlighted in bold.

CRITICAL ANALYSIS

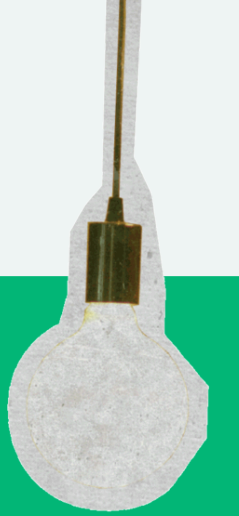
LIMITATIONS

- GradSafe's performance depends on the alignment quality of the base LLM (performs better with Llama-2 chat model which goes alignment than with Llama-2 base model which is not aligned).
- It provides binary safe/unsafe classification without fine-grained taxonomy of harmful content types.
- It is sensitive to the choice of reference prompts.
- Since it relies on gradient access, it cannot be applied to closed-source APIs like GPT-4.
- Since the source code and software are publicly available, adaptive attacks may be developed based on the detection results.



SUGGESTIONS FOR IMPROVEMENT

- Extending to multi-class classification to identify specific categories of unsafe content.
- Exploring gradient-based detection with other aligned LLMs to test robustness.
- Reducing dependency on a fixed compliance response by learning adaptive triggers.
- Investigating adversarial robustness, since attackers may design adaptive jailbreaks to evade gradient-based detectors.
- Combining gradient-based detection with textual or embedding-based methods for hybrid detection.



THANK

YOU